Structure and Interpretation of Deep Neural Networks

Hands-on Exercises

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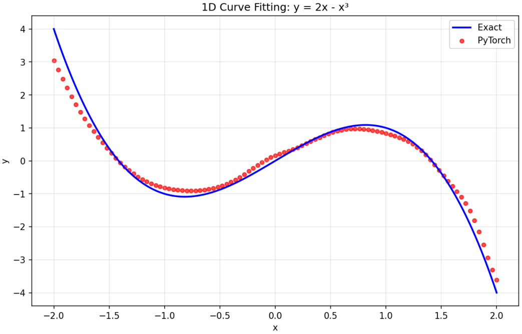
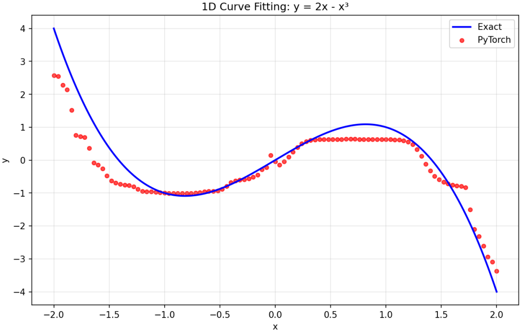
Hands-on exercise 5

- layer1 = nn.Linear(1, 10)
- layer2 = nn.Linear(10, 20)
- layer3 = nn.Linear(20, 10)
- layer4 = nn.Linear(10, 1)
- tanh(layer[1-3]), identity



Deep Network Training (Vanishing Gradients)

- layer1 = nn.Linear(1, 10)
- layer2 = nn.Linear(10, 20)
- layer3 = nn.Linear(20, 10)
- layer4 = nn.Linear(20, 20)
- layer5 = nn.Linear(20, 20)
- layer6 = nn.Linear(20, 10)
- layer7 = nn.Linear(10, 1)
- tanh(layer[1-6]), identity



$$x_1 \xleftarrow{f_1'} x_2 \xleftarrow{f_2'} x_3 \xleftarrow{f_3'} x_4 \xleftarrow{f_4'} x_5 \xleftarrow{f_5'} x_6 \xleftarrow{f_6'} x_7 \tag{1}$$

- Backpropagation via many layers gradient $\ll 1$ or ≈ 0
- Early layers learn very slowly or stop learning entirely
- The network fails to capture long-range dependencies

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O my mind, remember all that has been done. Remember – remember all that has been done.

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